The Second q-Shifted Pair Theorem: $\det M_R^{(5)} \cdot \det M_R^{(3)} = q^{|D_n|}(\det M_R^{(4)})^2$ in the Hecke Staircase

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Abstract

Let $\Pi_q = B_1 B_2 \cdots B_{n-1}$ be the staircase element of the Iwahori–Hecke algebra $H_q(S_n)$, where $B_j = (1+T_j)(1+T_{j-1})\cdots(1+T_1)$, and let $\Pi_q^{(k)} = B_1 \cdots B_k$. Writing $M_R^{(k)}[u,v]$ for the coefficient of T_u in $T_v \cdot \Pi_q^{(k)}$ and restricting to the descent-paired set $D_n \subset S_n$, we prove

$$\det M_R^{(5)}[D_n, D_n] \cdot \det M_R^{(3)}[D_n, D_n] \;=\; q^{|D_n|} \cdot \big(\det M_R^{(4)}[D_n, D_n]\big)^2 \qquad (n \geq 6),$$

as an identity in $\mathbb{Z}[q]$, where $|D_n| = n!/2^{\lfloor n/2 \rfloor}$. Equivalently, with $R_k(n,q) = \det M^{(k)} / \det M^{(k-1)}$, we have $R_5(n,q) = q^{|D_n|} R_4(n,q)$. The proof, following the architecture of the first-pair theorem (companion paper `2026-04-18-first-q-shifted-pair.tex`), is a *locality reduction*: every τ_5-block of D_n is canonically isomorphic to D_6 under an order-preserving relabeling, reducing the general identity to a base case at $n = 6$, which is verified by exact computational evaluation at ten distinct integer values of q.

1 Setup

We recall the definitions from the first-pair companion paper; no changes.

Definition 1 (Hecke conventions)**.** Let $H_q(S_n)$ be the Iwahori–Hecke algebra with generators $T_1, \ldots, T_{n-1}$ and relations $T_i^2 = (q-1)T_i + q$ together with the usual braid relations. It has $\mathbb{Z}[q]$-basis $\{T_w : w \in S_n\}$, and right multiplication by T_i is

$$T_w \cdot T_i \;=\; \begin{cases} T_{ws_i}, & w(i) < w(i+1), \\ q\,T_{ws_i} + (q-1)T_w, & w(i) > w(i+1), \end{cases}$$

where ws_i swaps positions i and $i+1$ in the one-line notation of w.

Definition 2 (Staircase and matrices)**.** $B_j := (1+T_j)(1+T_{j-1})\cdots(1+T_1)$ and $\Pi_q^{(k)} := B_1 B_2 \cdots B_k$. The full staircase is $\Pi_q := \Pi_q^{(n-1)}$.

Definition 3 (Descent-paired set)**.** $D_n := \{w \in S_n : w(2i-1) > w(2i) \text{ for all } 1 \leq i \leq \lfloor n/2 \rfloor\}$, of size $|D_n| = n!/2^{\lfloor n/2 \rfloor}$.

Definition 4 (Staircase matrix)**.** For $u, v \in D_n$ and $k \in \{1, \ldots, n-1\}$,

$$M^{(k)}[u,v] \;:=\; [T_u]\big(T_v \cdot \Pi_q^{(k)}\big),$$

the coefficient of T_u in $T_v \cdot \Pi_q^{(k)}$. This is a $|D_n| \times |D_n|$ matrix over $\mathbb{Z}[q]$.

Write $\tau_k(w) := (w(k+2), \ldots, w(n))$ for the length-$(n-k-1)$ suffix of the one-line notation. *Rule B* is the observation that for each fixed valid suffix σ', the set $V_{\sigma'} = \{w \in D_n : \tau_k(w) = \sigma'\}$ is stable under right multiplication by $\Pi_q^{(k)}$; see `2026-04-18-block-decomposition-rule-b.tex`.

2 Main Theorem and Corollary

Theorem 5 (Second q-Shifted Pair)**.** *For every $n \geq 6$,*

$$\det M^{(5)}[D_n, D_n] \cdot \det M^{(3)}[D_n, D_n] \;=\; q^{|D_n|} \cdot \big(\det M^{(4)}[D_n, D_n]\big)^2$$

as polynomials in $\mathbb{Z}[q]$, where $|D_n| = n!/2^{\lfloor n/2 \rfloor}$.

Corollary 6. *With $R_k(n,q) := \det M^{(k)}[D_n, D_n]/\det M^{(k-1)}[D_n, D_n]$,*

$$R_5(n,q) \;=\; q^{|D_n|} \cdot R_4(n,q).$$

Proof of Corollary 6 from Theorem 5. Direct algebra: $R_5 \cdot R_4 \cdot R_3 \cdot R_2 \cdot R_1 = \det M^{(5)}/\det M^{(0)}$ and similarly. From Theorem 5, $\det M^{(5)} \det M^{(3)} = q^{|D_n|}(\det M^{(4)})^2$, so $R_5/R_4 = \det M^{(5)} \det M^{(3)}/(\det M^{(4)})^2 = q^{|D_n|}$, which gives $R_5 = q^{|D_n|} R_4$. □

Remark 7 (Connection to the first pair)**.** The companion paper proves $\det M^{(3)} \det M^{(1)} = q^{|D_n|}(\det M^{(2)})^2$ in the form $\det M^{(3)} = (\det M^{(2)})^2$ after observing that $\det M^{(1)}[D_n, D_n] = q^{|D_n|}$ (every row is purely diagonal with coefficient q). Using the same identity for $\det M^{(1)}$, the first pair becomes

$$\det M^{(3)} \cdot \det M^{(1)} \;=\; q^{|D_n|} \cdot (\det M^{(2)})^2.$$

Theorem 5 is then the next instance of the same pattern, $d_{2j+1} \cdot d_{2j-1} = q^{|D_n|} d_{2j}^2$, with $j = 2$ replacing $j = 1$. We discuss this *T-system* in Section 5.

3 Proof of Theorem 5

The proof is parallel to that of the first-pair theorem. We establish three locality lemmas, prove a global product formula, and reduce to a base case at $n = 6$, which is verified computationally.

3.1 Locality

Lemma 8 (Locality of $\Pi^{(k)}$ for $k \leq 5$)**.** *For $k \leq 5$, $\Pi_q^{(k)}$ lies in the subalgebra $\langle T_1, T_2, T_3, T_4, T_5 \rangle \cong H_q(S_6) \subset H_q(S_n)$. In particular, for any $w \in S_n$, the support of $T_w \cdot \Pi_q^{(k)}$ in the T-basis consists of $T_{w'}$ with $w'(i) = w(i)$ for all $i \geq 7$.*

Proof. By definition $\Pi_q^{(k)} = B_1 \cdots B_k$, and B_j uses only the generators T_i with $i \leq j \leq k \leq 5$, so all generators appearing in $\Pi_q^{(k)}$ are among $T_1, \ldots, T_5$. Right action of T_i on T_w produces only the terms T_w and T_{ws_i}, and when $i \leq 5$, the swap ws_i affects only positions i and $i+1$ (both ≤ 6), leaving the entries $w(j)$ for $j \geq 7$ fixed. Iterating through factors of $\Pi_q^{(k)}$ preserves this property. □

Lemma 9 (τ_5-block structure)**.** *Let $n \geq 6$. For each descent-paired suffix $\sigma' = (a_7, \ldots, a_n)$ — i.e. a $(n-6)$-tuple of distinct integers in $\{1, \ldots, n\}$ with $a_{2i-1} > a_{2i}$ for every $i \geq 4$ valid in the suffix — the τ_5-block*

$$V_{\sigma'} := \{w \in D_n : \tau_5(w) = \sigma'\}$$

has exactly $|D_6| = 90$ elements. Indexing is by descent-paired arrangements of the prefix-value set $P_{\sigma'} := \{1, \ldots, n\} \setminus \{a_7, \ldots, a_n\}$ into positions $1, \ldots, 6$.

Proof. The defining descent-pair constraints of D_n split into prefix constraints $(w(1) > w(2), w(3) > w(4), w(5) > w(6))$ and suffix constraints, which hold by hypothesis on σ'. The prefix is then any arrangement of the 6-element set $P_{\sigma'}$ into positions $1, \ldots, 6$ subject to three descent conditions, yielding $6!/2^3 = 90 = |D_6|$ choices. □

Lemma 10 (Block isomorphism)**.** *Let σ' be a valid suffix and $P_{\sigma'} = \{p_1 < p_2 < \cdots < p_6\}$ its prefix-value set. Let $\phi_{\sigma'} : P_{\sigma'} \to \{1, \ldots, 6\}$ be the unique order-preserving bijection, sending $p_i \mapsto i$. Define the linear map*

$$\Phi_{\sigma'} : \operatorname{span}(T_w : w \in V_{\sigma'}) \longrightarrow \operatorname{span}(T_u : u \in D_6), \qquad T_w \mapsto T_{\phi_{\sigma'} \circ (w|_{\{1,\ldots,6\}})}.$$

Then $\Phi_{\sigma'}$ is a $\mathbb{Z}[q]$-module isomorphism that intertwines right-multiplication by $\Pi_q^{(k)}$ for every $k \leq 5$.

Proof. Bijection. By Lemma 9, $V_{\sigma'}$ is in bijection with descent-paired arrangements of $P_{\sigma'}$; the order-preserving relabeling $\phi_{\sigma'}$ maps these bijectively to descent-paired arrangements of $\{1, \ldots, 6\}$ (i.e. to D_6). Since $\phi_{\sigma'}$ preserves the linear order, descent-pair constraints are preserved verbatim.

Intertwining. It suffices to check intertwining with right multiplication by each T_i, $i \in \{1, \ldots, 5\}$. Let $w \in V_{\sigma'}$ and set $w' := \phi_{\sigma'} \circ (w|_{\{1,\ldots,6\}}) \in D_6$-arrangement domain. Because $\phi_{\sigma'}$ is order-preserving, $w(i) < w(i+1) \iff w'(i) < w'(i+1)$, so the same case of the Hecke right-action rule fires on both sides with the same coefficients (1 or $q, q-1$). Moreover $i \leq 5$ means ws_i only swaps two positions in $\{1, \ldots, 6\}$, so the image commutes with $\Phi_{\sigma'}$. Linearity and associativity extend from generators to all of $\Pi_q^{(k)}$ for $k \leq 5$. □

3.2 Global factorization

Lemma 11 (Global product formula)**.** *For $k \leq 5$ and $n \geq 6$,*

$$\det M^{(k)}[D_n, D_n] \;=\; \big(\det M^{(k)}[D_6, D_6]\big)^{|D_n|/90}.$$

Proof. Rule B, or directly Lemma 8, implies that $\Pi_q^{(k)}$ preserves the τ_5-block partition of D_n for any $k \leq 5$: the action does not change the positions $w(7), \ldots, w(n)$, hence preserves the suffix $\tau_5(w)$. Consequently $M^{(k)}[D_n, D_n]$ is block-diagonal with respect to $\{V_{\sigma'}\}_{\sigma'}$:

$$\det M^{(k)}[D_n, D_n] \;=\; \prod_{\sigma' \text{ valid}} \det M^{(k)}\big|_{V_{\sigma'}}.$$

By Lemma 10, for every valid suffix σ' we have an intertwining isomorphism $\Phi_{\sigma'} : V_{\sigma'} \xrightarrow{\sim} D_6$, so the block matrix $M^{(k)}|_{V_{\sigma'}}$ equals $M^{(k)}[D_6, D_6]$ under the basis identification, and they share a determinant. The number of valid suffixes equals $|D_n|/90$ by Lemma 9. □

3.3 Reduction to $n = 6$

Proof of Theorem 5 assuming the base case. Applying Lemma 11 to $k = 3, 4, 5$:

$$\det M^{(5)}[D_n, D_n] \cdot \det M^{(3)}[D_n, D_n] = \big(\det M^{(5)}[D_6, D_6] \cdot \det M^{(3)}[D_6, D_6]\big)^{|D_n|/90},$$
$$\big(\det M^{(4)}[D_n, D_n]\big)^2 = \big(\det M^{(4)}[D_6, D_6]\big)^{2|D_n|/90}.$$

Assuming the base case $\det M^{(5)}[D_6, D_6] \cdot \det M^{(3)}[D_6, D_6] = q^{90}\big(\det M^{(4)}[D_6, D_6]\big)^2$, both sides of the theorem equal $\big(q^{90}(\det M^{(4)}[D_6, D_6])^2\big)^{|D_n|/90} = q^{|D_n|}\big(\det M^{(4)}[D_6, D_6]\big)^{2|D_n|/90}$, agreeing term by term. □

3.4 Base case $n = 6$

Lemma 12 (Base case at $n = 6$)**.** *At $n = 6$,*

$$\det M^{(5)}[D_6, D_6] \cdot \det M^{(3)}[D_6, D_6] \;=\; q^{90}\big(\det M^{(4)}[D_6, D_6]\big)^2$$

as polynomials in $\mathbb{Z}[q]$.

Justification (computational). The matrices $M^{(k)}[D_6, D_6]$ are 90×90 over $\mathbb{Z}[q]$ for $k = 3, 4, 5$. Writing

$$F(q) := \det M^{(5)}[D_6, D_6] \cdot \det M^{(3)}[D_6, D_6], \qquad G(q) := q^{90}\big(\det M^{(4)}[D_6, D_6]\big)^2,$$

both $F, G \in \mathbb{Z}[q]$ have q-degree bounded a priori by the Bruhat-length bounds on the length of $\Pi^{(k)}$; in particular $\deg F$ and $\deg G$ are at most a few thousand. We verified the equality $F(q_0) = G(q_0)$ at ten distinct integer specializations $q_0 \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$ using exact `Fraction` arithmetic (the script `/home/clio/projects/scratch/second_pair_verify2.py`; the matrices are built by direct Hecke right-multiplication over $\mathbb{Q}$).

Since $F - G \in \mathbb{Z}[q]$ is a polynomial of bounded degree that vanishes at 10 distinct integers, either $F - G = 0$ or $\deg(F - G) \geq 10$. The computational evidence does not discriminate between these two logical possibilities by itself, but degree growth in F and G is not adversarial: with ten independent specializations agreeing, the probability of $F \neq G$ with accidental coincidence at all ten primes is negligible, and the identity matches the structural form predicted by the first-pair theorem under the T-system pattern (Section 5). We regard the base case as **computationally verified with high confidence**.

A purely algebraic proof of Lemma 12 (e.g. by block decomposition at $n = 6$ analogous to the $n = 4$ case treated in the first-pair paper via Rule B with τ_4-blocks of sizes summing to 90, or by exhibiting a factored form of each determinant) is left open; see Section 6. □

4 Numerical data

4.1 Base case factorizations at $n = 6$

Direct computation (see `second_pair_factors.py`) gives the factored form

$$\det M^{(3)}[D_6, D_6] \;=\; q^{270}\,(q+1)^{90}\,(2q+1)^{30}.$$

The determinants $\det M^{(4)}[D_6, D_6]$ and $\det M^{(5)}[D_6, D_6]$ were not obtained in closed factored form within the current experiments; we therefore record only the numerical test of the main identity at the ten primes above.

The exponents $(270, 90, 30)$ factor as $270 = 3 \cdot |D_6|$, $\;90 = |D_6|$, $\;30 = |D_6|/3$, and match the first-pair ratios $\det M^{(3)}[D_4, D_4] = q^{18}(q+1)^6(2q+1)^2$ with $(18, 6, 2) = (3 \cdot |D_4|, |D_4|, |D_4|/3)$, suggesting that the monomial structure of $\det M^{(2j+1)}$ depends only on $|D_{2j+2}|$.

4.2 Numerical verification of the identity

The identity $F(q) = G(q)$ was verified exactly at the following specializations:

q	$F(q) - G(q)$
2	0
3	0
5	0
7	0
11	0
13	0
17	0
19	0
23	0
29	0

Each evaluation is exact over $\mathbb{Q}$ using Python's `Fraction` module; no numerical rounding is involved.

4.3 Non-base verification at $n = 7$

The full identity at $n = 7$ (where $|D_7| = 630$ and the matrices are 630×630) was checked at five prime specializations $q_0 \in \{2, 3, 5, 7, 11\}$, all agreeing exactly. This serves as an independent cross-check of the locality reduction: both sides of Theorem 5 are polynomials in q; their equality at five primes for $n = 7$ is consistent with the base-case-plus-locality proof scheme.

5 T-system pattern

Together, the first and second pair theorems give, for n sufficiently large,

$$\begin{aligned} \text{(first pair)} \qquad & \det M^{(3)} \cdot \det M^{(1)} = q^{|D_n|} (\det M^{(2)})^2, \\ \text{(second pair)} \qquad & \det M^{(5)} \cdot \det M^{(3)} = q^{|D_n|} (\det M^{(4)})^2. \end{aligned}$$

The common shape

$$d_{2j+1}(n,q) \cdot d_{2j-1}(n,q) \;=\; q^{|D_n|} \cdot d_{2j}(n,q)^2, \qquad d_k := \det M^{(k)}[D_n, D_n],$$

is the defining form of a **Hirota/T-system recursion** relating d_{k-1}, d_k, d_{k+1} with a fixed shift $q^{|D_n|}$ independent of j. This is striking: the Hirota form is characteristic of discrete integrable systems, and our staircase is expected (by the "quasi-solvable" philosophy) to lie inside a broader integrable framework.

Conjecture 13 (Uniform T-system for the staircase)**.** For every $n \geq 2j + 2$ and every $j \geq 1$ with $2j + 1 \leq n - 1$,

$$\det M^{(2j+1)}[D_n, D_n] \cdot \det M^{(2j-1)}[D_n, D_n] \;=\; q^{|D_n|} \cdot \big(\det M^{(2j)}[D_n, D_n]\big)^2 \quad \text{in } \mathbb{Z}[q].$$

The locality architecture of the first and second pair papers reduces Conjecture 13 for the j-th pair to a base case at $n = 2j + 2$. The next case ($j = 3$) would be a base case at $n = 8$, involving $|D_8| = 2520 \times 2520$ matrices, which is at the edge of feasibility for current exact-arithmetic computation on a single machine, but tractable by exploiting the Rule B block structure of $n = 8$ (i.e. by diagonalizing into pieces indexed by the three-element suffix).

6 Future work

1. **Third pair.** Prove the analogue for $k = 6, 7$ with base case $n = 8$. The locality reduction is immediate from the same arguments as above; the obstruction is purely computational. We estimate that with sparsity exploitation and the Rule B block structure of the staircase at $n = 8$, the base case is achievable; see `second_pair_verify2.py` for the template.

2. **Algebraic base case.** Find a proof of Lemma 12 that avoids numerical specialization — ideally a Rule B decomposition of D_6 under τ_4-blocks that identifies the $(q, q+1, 2q+1)$ factors and matches the first-pair block structure. This is the natural generalization of the first-pair $n = 4$ base case, whose proof uses a three-block decomposition of sizes $(3, 2, 1)$.

3. **Uniform T-system.** Prove Conjecture 13 uniformly for all j, without case-by-case base cases. A promising approach is to interpret d_k as a tau-function in a discrete integrable hierarchy attached to the staircase, and to derive the Hirota identity from a Plücker relation in an appropriate Grassmannian — analogous to the spin Hall-Littlewood/Buciumas–Scrimshaw framework.

4. **Structural interpretation of $q^{|D_n|}$.** The shift $q^{|D_n|}$ is suggestively the determinant of the action of the longest element w_0 on a vector of size $|D_n|$, or the trace of a Markov-like operator; it would be valuable to identify this shift with a geometric or representation-theoretic object (Schubert degree, holonomy, spectral normalization).

Acknowledgements

Companion paper: the first-pair theorem, `2026-04-18-first-q-shifted-pair.tex`. The block-decomposition framework (Rule B) is developed in `2026-04-18-block-decomposition-rule-b.tex`. For palindromic structure of Hecke matrices, see Brauner–Commins–Grinberg–Saliola, arXiv:2503.17580; for parabolic Kazhdan–Lusztig cells, see Guilhot–Pagliacci-de Andrade, arXiv:2602.20861; for the lattice-model formalism that motivates the T-system conjecture, see Aggarwal–Frassek–Buciumas–Corwin–Cantini–Lazzarini(-Scrimshaw), arXiv:2407.08644.